

ZIQI WEN

Email: ziqiwen@ucsb.edu **Webpage:** <https://starsky77.github.io/>

EDUCATION

University of California, Santa Barbara, Santa Barbara, CA
Doctor of Philosophy in Computer Science | **GPA:** 4.0/4.0 Sept. 2024 - May 2029
Research Interests: Scene Understanding, Visual Cognition
Advisor: Prof. Miguel P. Eckstein

Carnegie Mellon University - School of Computer Science, Pittsburgh, PA
Master of Computational Data Science | **GPA:** 3.87/4.0 Aug. 2022 - May 2024
Selected Coursework: Large Language Models, Image Synthesis, Deep Learning System, Cloud Computing

Zhejiang University Hangzhou, China
Bachelor of Engineering in Computer Science and Technology | **GPA:** 3.87/4.0 Aug. 2018 - Jun. 2022
Minor in Psychology | **Minor GPA:** 4.0/4.0

Imperial College London Remote
Data Science Summer School Jul. 2020 - Aug. 2020

PUBLICATIONS & PREPRINTS

Ziqi Wen, Parsa Madinei, Miguel P. Eckstein. *Revealing the Gap in Human and VLM Scene Perception through Counterfactual Semantic Saliency*. NeurIPS 2026.

Shravan Murlidaran, **Ziqi Wen**, Sana Shehabi, Miguel P. Eckstein. *Why We Look Where We Look: Emergent Human-like Fixations of a Foveated Visual Language Model Maximizing Scene Understanding*. arXiv preprint arXiv:2605.17823 (2026).

Ziqi Wen, Jonathan Skaza, Shravan Murlidaran, William Y. Wang, Miguel P. Eckstein. *Predicting Reaction Time to Comprehend Scenes with Foveated Scene Understanding Maps*. arXiv preprint arXiv:2505.12660 (2025).

Parsa Madinei, Ryan Solgi, **Ziqi Wen**, Jonathan Skaza, Miguel P. Eckstein, Ramtin Pedarsani. *INTERLACE: Interleaved Layer Pruning and Efficient Adaptation in Large Vision-Language Models*. CVPR 2026 Findings.

Jonathan Skaza, Parsa Madinei, **Ziqi Wen**, Miguel P. Eckstein. *DReX: Pure Vision Fusion of Self-Supervised and Convolutional Representations for Image Complexity Prediction*. arXiv preprint arXiv:2511.16991 (2025).

Tianqin Li, **Ziqi Wen**, Leiran Song, Jun Liu, Zhi Jing, Tai Sing Lee. *From Local Cues to Global Percepts: Emergent Gestalt Organization in Self-Supervised Vision Models*. arXiv preprint arXiv:2506.00718 (2025).

Jonathan Skaza, Shravan Murlidaran, Apurv Varshney, **Ziqi Wen**, William Y. Wang, Miguel P. Eckstein, Michael Beyeler. *A Deep Learning Framework for Predicting Functional Visual Performance in Bionic Eye Users*. bioRxiv (2025).

Ziqi Wen, Tianqin Li, Tai Sing Lee. *Does resistance to style-transfer equal Global Shape Bias? Measuring network sensitivity to global shape configuration*. ICLR 2024 Workshop Re-Align.

Tianqin Li, **Ziqi Wen**, Yangfan Li, Tai Sing Lee. *Emergence of Shape Bias in Convolutional Neural Networks through Activation Sparsity*. NeurIPS 2023(Oral).

PRESENTATIONS & TALKS

Ziqi Wen, Jonathan Skaza, Shravan Murlidaran, Miguel P. Eckstein. *Contributions of Foveation, Scene Ambiguity, and Visual Complexity To Human Time to Comprehend Scenes*. **Talk** presented at the Vision Sciences Society (VSS) Annual Meeting, 2026.

Ziqi Wen, Jonathan Skaza, Shravan Murlidaran, William Y. Wang, Miguel P. Eckstein. *Foveated Multi-Modal Large Language Model Maps to Predict Time to Understand Scenes*. Poster presented at the Vision Sciences Society (VSS) Annual Meeting, 2025.

HONORS & AWARDS

VSS 2026 Talk

NeurIPS 2023 Oral

Outstanding Graduates of Zhejiang University, 2022

Outstanding Graduation Project of Zhejiang University, 2022

TEACHING EXPERIENCE

University of California, Santa Barbara

CS5A: Intro to Data Science

CS181A: Intro to Computer Vision

CS190A: Foundation of Machine Learning

Teaching Assistant

Spring 2026

Fall 2024, Winter 2026

Spring 2025

Carnegie Mellon University

15386/686: Neural Computation

Teaching Assistant

Spring 2024

RESEARCH EXPERIENCE

Center for the Neural Basis of Cognition & Computer Science Department, Carnegie Mellon University

Shape and texture bias in computer vision models and their benefits

Feb. 2023 - May 2024

Supervisor: Prof. Tai Sing Lee

- **Emergence of Shape Bias in Convolutional Neural Networks through Activation Sparsity**
 - Enforcing the sparse coding constraint using a non-differential Top-K operation can lead to the emergence of structural encoding in neurons in convolutional neural networks.
 - The emergence of shape bias benefits for different network structures with various datasets on different tasks. (e.g. object recognition, image synthesis)
 - Accepted as **NeurIPS 2023(oral)** (top 2%)
- **Does resistance to style-transfer equal Global Shape Bias? Measuring network sensitivity to global shape configuration**
 - Show that stylized trained neural network still focus on local feature rather than global shape.
 - Provide Distorted Shape Testbench as an alternative measurement of global shape sensitivity, evaluate both human and multiple deep learning models.
 - Accepted as **ICLR 2024 Workshop Re-Align** poster

Human-Computer Interaction Institute, Carnegie Mellon University

Analysis of Online Interpersonal Conflict

Apr. 2023 - Dec. 2023

Supervisor: Prof. Robert E. Kraut & Prof. John M. Levine

- Analysis how interpersonal conflict influence the consequent behavior of the users in Wikipedia Talk Page and their participation in the conversations based on the WiKiDetox dataset.

State Key Laboratory of CAD & CG, Zhejiang University

Efficient Neighbor Gathering Methods for Large-scale Point Clouds

Apr. 2021 - Dec. 2021

Supervisor: Prof. Zhaopeng Cui

- Optimize the neighbor gathering in Dynamic Graph CNN with One-Shot Neural Architecture Search(NAS) and efficient neighbors gathering methods.
- Speeds up the baseline **4 times** and reduces memory cost by **34%** with similar accuracy in the same testing condition. Enlarge the maximum processing capacity of baseline by **20 times**, able to process near million points.